

Executive Summary: Customer Retention Analysis

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Data Context (via Kaggle / UCI)

The Online Retail II data set contains all the transactions occurring for a UK-based and registered, non-store online retail between 01/12/2009 and 09/12/2011. The company mainly sells unique all-occasion gift-ware. Many customers of the company are wholesalers.

Objective

Analyze customer behavior using online retail shopping data (Online Retail II, 2009–2011) to understand retention patterns by cohort and identify business opportunities.

Key Findings

- **Steep early drop-off:** Month to month repeat customer retention falls to around 20% of the original onboarding cohort after 1 month from first purchase. Meaning four out of five customers do not return after their first purchase the following month.
- **Flat long tail:** After Month 3, retention stabilizes at 15–20% of the original cohort, this is seen across all cohorts. A small loyal core continues to buy regularly.
- **Consistent patterns:** Retention curves are nearly identical across time. Newer cohorts (2010–2011) did not retain better than older ones.
- **Cohort size trend:** The largest cohort was Dec 2009 (~955 customers) and had a higher retention. Later cohorts were smaller (300–400 customers) and showed no improvement in loyalty between them.
- **Revenue growth driver:** Rising revenue and active users during 2010–2011 were powered by new acquisitions, not by stronger retention.

Business Implications

- The current growth model is acquisition-led, but long-term value is undermined by poor retention.
- The most important window for retention is the 1st month. Customers who return once are much more likely to become part of the loyal base.
- Marketing dollars may be better spent on programs to encourage quick and often repeat business, rather than purely on acquisition.

Recommendations

1. **Strengthen onboarding:** Welcome campaigns, personalized offers, and reminders within the first 30 days.
2. **Trigger re-engagement:** Target inactive customers after their first purchase with promotions or outreach.
3. **Segment loyal base:** Identify the ~15% who stay beyond Month 3, and explore cross-sell/up-sell opportunities.
4. **Update cohort metric data often:** Use dashboards to monitor whether new strategies shift the retention curves upward.

Deliverables

- **SQL Pipeline (DuckDB):** Cleaned transactional data and retention tables.
- **Tableau Dashboard:** Interactive heatmap, retention curves, and KPI trends.
- **Insights Summary:** Cohort retention analysis highlighting Month 1 churn as the key driver of lifetime value.